DEPARTMENT OF MATHEMATICS & STATISTICS
MTH 211: THEORY OF STATISTICS
Session: 2025-26 Semester: (II)

Quiz 1

Maximum Marks: 20 Time: 50 minutes

1. Let $X_1, \ldots, X_n$ be a random sample from the $U(\theta, \theta+1), \theta \in \Re$, distribution.

(a) Find a statistic that is minimal sufficient for θ.

(b) Prove or disprove that the minimal sufficient obtained in (a) is complete.

(2 + 4 = 6 marks)

2. Let $X_1, \ldots, X_n$ be a random sample from the $N(\theta, \theta), \theta > 0$ distribution.

(a) Verify whether a minimal sufficient statistic is complete.

(b) Based on the entire sample, obtain an unbiased estimator of $g(\theta) = \theta^2$.

(3 + 3 = 6 marks)

3. Let $X_1, \ldots, X_n$ be a random sample from a Geometric distribution with pmf:
$P(X = x) = (1-p)^{(x-1)}p, \quad x = 1, 2, \ldots; \quad 0 < p < 1$
We wish to estimate $g(p)$ = probability of observing a "success" on the first trial.

(a) Find a complete sufficient statistic for $g(p)$.

(b) Calculate the UMVUE for $g(p)$

(3 + 5 = 8 marks)